

Zurich ServiceNow AI Platform Capabilities

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CI relationships in the CMDB

The CMDB, in contrast to a static asset list, helps you track not only the configuration items (CIs) within your system, but also the relationships between those items.

A relationship in the CMDB consists of two CIs and a relationship type:

- Parent CI
- Child CI
- Type of the relationship that links both CIs

For example, in the [Server1] [Managed by] [Server2] relationship:

- Server1 is the child CI
- Server2 is the parent CI
- [Managed by] is the relationship type

For example, a web application might read data from an instance of Oracle, which in turn might depend on a piece of underlying hardware. Most CIs in a CMDB have multiple relationships to other CIs, users, and groups.

The relationships between CIs can be automatically discovered. If you use Discovery, many relationships can be automatically loaded into the system through the discovery process. If you import your data from another system, you get some form of relationships.

You can add to automatically discovered relationships, create relationships, or edit relationships for a CI by launching the [CI relationship editor](#) from the CI form. As an alternative to the CI relationship editor, [Unified Map](#) in the [CMDB Workspace store app](#) provides the latest functionality for editing CI relationships. For more information, see [Edit relationships in Unified Map](#).

Dependent and non-dependent relationships

[Dependent relationships](#), such as tomcat RunsOn Hardware, are used by the Identification and Reconciliation Engine (IRE) to identify dependent

CIs. The IRE avoids duplicate entries of CIs into your Configuration Management Database (CMDB) by leveraging these relationships to determine if a recently discovered CI is already in the CMDB.

For non-dependent relationships, the CMDB tracks the discovery source and the last scanned time in the Relationship Sources [sys_rel_source] table. Non-dependent relationships aren't used for CI identification, and can be deleted when no longer needed.

To avoid burdening your IRE with excessive load, by default, the Relationship Sources [sys_rel_source] table doesn't auto-populate. If you want to track full information on non-dependent relationships, you can change the default using the glide.identification_engine.populate_sys_rel_source property.

Dependent relationships are used for CI identification, so they shouldn't be directly deleted as they aren't tracked.

Information in the Relationship Sources [sys_rel_source] table can be used to decide if it's safe to delete a potentially non-dependent relationship. For example, a discovery source, which is attempting to delete a non-dependent relationship can confirm that:

- There are no other data sources for that relationship.
- The relationship wasn't updated for some specified length of time and therefore is no longer needed.

When a non-dependent relationship is deleted from the CI Relationship [cmdb_rel_ci] table, all cascading corresponding records in the Relationship Sources [sys_rel_source] table are deleted.

Key relationships

The following table contains descriptions for some key CMDB relationships.

Parent	Child	Description
Applicative Flow To	Applicative Flow From	Connections between endpoint CIs.

Parent	Child	Description
		Note: For internal use only (service model).
Connects to	Connected by	Network Connections between elements that are talking to each other. Examples: Workstation to switch, switch to switch, kubernetes workload to service.
Contains	Contained by	Typically a containment relationship (CI to contained CI). The child CI typically has a single parent CI with this relationship type. Examples: Tomcat to Tomcat WAR, VMware Datacenter contains Network.
Defines resources for	Gets resources from	Parent CI defines/gets resources from a child CI. Example: VMware - Resource pool gets

Parent	Child	Description
		resources from ESX Server.
Depends on	Used by	Parent CI depends on child CI. Meaning that problem/change in the child CI may impact the parent CI.
Hosted on	Hosts	Hosting relationship between an element and its host. Examples: Cloud resource to logical data center, k8s workload to k8s cluster.
Implement End Point To	Implement End Point From	Endpoint to CI that exposes this endpoint. Note: For internal use only (service model).
Manages	Managed by	Typically used where one CI manages one or more other CIs. Example: vCenter manages vCenter Datacenter.

Parent	Child	Description
Members	Member of	Typically used with clusters where a cluster node is a member of a cluster. Example: ESXi Server is a member of vCenter Cluster.
Owns	Owned by	Usually a containment relationship (CI to owned CI). The child CI typically has a single parent with this relationship type.
Runs on	Runs	Typically between a CI that represents a software application, to the hosting hardware/VM. Example: Tomcat 'Runs on' Linux server.
Use End Point To	Use End Point From	From the CI to an outgoing endpoint. Note: For internal use only (service model).

- Suggested class relationships

The system keeps a table (Suggested Relationship [cmdb_rel_type_suggest]) of relationship types that are appropriate for a CI type, based on its class. You can manage suggested relationships by navigating to **Configuration > Suggested Relationships**, or in the CI Class Manager.

- [Add a suggested relationship](#)

Add a suggested relationship for a class. The list of suggested relationships for a class is available when you create a new relationship for a CI of that class.

- [Relationship governance rules](#)

Relationship governance rules is a set of relationship rules used to ensure consistency and validity in modeling relationships between configuration items (CIs) in the CMDB. Use relationship governance rules to prevent the selection of relationship types or directions that are not allowed between specific CI types.

- [CI relations formatter](#)

The default CI form includes a CI relations formatter from which you can examine a CI and its relationships in various views. From the CI relations formatter, you can also launch the CI relationship editor for the CI.

- [CI relationship editor](#)

Use the relationship editor to view, create, modify, or delete CI relationships. Open the relationship editor from the CI Relations formatter.

- [Relation qualifier](#)

A relation qualifier, which is a CI of the Qualifier [cmdb_ci_qualifier] type, stores important information about the CI relationships.

- [CI relationship security](#)

When applying security to CI relationships, it is important to apply the access controls both to the CI Relationship (cmdb_rel_ci) table and to create an operation editCIRelations to the * table as well.

- [Create a CI relation rollup](#)

A CI relation rollup allows you to sum, count, max, min, or mean a relationship type. You can create CI relation rollups.

Related tasks

- [Create a CI relation rollup](#)

Suggested class relationships

The system keeps a table (Suggested Relationship [cmdb_rel_type_suggest]) of relationship types that are appropriate for a CI type, based on its class. You can manage suggested relationships by navigating to **Configuration > Suggested Relationships**, or in the CI Class Manager.

Suggestion model

The relationship editor has a base CI. The base CI designates the CI that a user was on before launching the editor, as the base CI in the new relationship. If you launched the relationship editor from the Inux100 CI, then Inux100 becomes the base CI. Also, every CI in the system has a type (class). For example, bond Inux100 is of the Linux server type.

Many CI types are children of other types in the hierarchy. For example, the class hierarchy for a Linux server is:

```
cmdb_ci -> cmdb_ci_hardware -> cmdb_ci_computer ->
cmdb_ci_server -> cmdb_ci_linux_server
```

The suggestion model works by analyzing the suggested relationship table for all relationships whose base class is the current base class or any one of its parent classes. For example, looking at a Linux server, the suggestion model would retrieve any relationships whose base class was:

```
cmdb_ci_linux_server, cmdb_ci_server, cmdb_ci_computer,
cmdb_ci_hardware, or cmdb_ci
```

ITOM Visibility, if available, uses enhanced discovery patterns to identify and add CI relationships to the Suggested Relationships table.

Suggested CI relationships in the relationship editor

The CI relationship editor uses the suggestion model to help users select reasonable relationships for configuration items.

For example, consider these relationship types in the system:

- Provides Power for :: Receives Power From
- Runs on :: Hosts

Typically, a user uses these relationships to define the following reasonable relationships between two items as follows:

- a database runs on a server
- a rack provides power for a server

Typically, neither of the following definitions would be appropriate:

- a rack runs on a server
- a server runs on a database

For descriptions of some key relationships, see [CI relationships in the CMDB](#).

Related tasks

- [Add a suggested relationship](#)
- [Create a CI relation rollup](#)
- [CI Class Manager](#)

Related concepts

- [Relationship governance rules](#)
- [CI relations formatter](#)
- [CI relationship editor](#)
- [Relation qualifier](#)

- CI relationship security

Add a suggested relationship

Add a suggested relationship for a class. The list of suggested relationships for a class is available when you create a new relationship for a CI of that class.

Before you begin

Role required: To view — itil. To create, update, or delete suggested relationships — itil_admin.

Procedure

1. Use the CI Class Manager (Role required: itil_admin):
 - a. Navigate to **All > Configuration > CI Class Manager**.
 - b. Click **Hierarchy** to expand the CI Classes list. Then select the class to add a suggested relationship to.
 - c. In the class navigation bar, click **Suggested Relationships**.
 - d. Click **New**.
 - e. In the Add Suggested Relationship dialog box, select a **Relationship** and a **Target Class** for the relationship. **This Class** and the **Target Class** become parent or child in the suggested relationship, based on your selection of the **Relationship**.
 - f. Click **Save**.
2. Or, navigate to **All > Configuration > Relationships > Suggested Relationships** (Role required: admin):
 - a. Click **New**.
 - b. Complete the form.

Suggested Relationship fields

Field	Description
Base class	The base class in the relationship, which depending on the relationship type, is either the parent or the child in the relationship.
Relationship	Relationship type.
Dependent class	The dependent class in the relationship, which depending on the relationship type, is either the parent or the child in the relationship.

Example: Suggested relationship you can add

Base Class	Relationship	Dependent/Target Class
Oracle	Is Hosted On	Linux Server
Oracle	Is Hosted On	Solaris Server

Note: The same parent class and relationship can appear more than once.

What to do next

You may need to delete a suggested relationship, for example, to limit the choice of available relationships in the CI relationship editor. Removing a suggested relationship does not affect relationships that are created or updated by Discovery.

Related tasks

- [Create a CI relation rollout](#)

Related concepts

- [Relationship governance rules](#)
- [CI relations formatter](#)
- [CI relationship editor](#)
- [Relation qualifier](#)
- [CI relationship security](#)

Related reference

- [Suggested class relationships](#)

Relationship governance rules

Relationship governance rules is a set of relationship rules used to ensure consistency and validity in modeling relationships between configuration items (CIs) in the CMDB. Use relationship governance rules to prevent the selection of relationship types or directions that are not allowed between specific CI types.

Different applications such as Discovery and Service Mapping create relationships between CIs. Each application might use inconsistent relationship type or direction to represent the same entity, resulting in multiple views of the same CIs. Relationship governance rules define what are valid relationship types and valid directions between pairs of CI types resulting in valid and consistent relationships in the CMDB.

Relationship governance rules consist of:

- **CMDB dependent relationship rules:** Rules (hosting and containment rules) that are used for CI identification. You can view and modify dependent relationship rules in the CI Class Manager, after selecting a class from the class hierarchy and clicking **Dependent Relationship**.
- **Suggested relationships:** Rules that are based on existing suggested relationships in the Suggested Relationship [cmdb_rel_type_suggest] table. Suggested relationships are used in the [CI relationship editor](#). You can view and modify suggested relationships in the CI Class Manager,

after selecting a class from the class hierarchy and clicking **Suggested Relationships**.

- **Reference rules:** Rules that are used mostly by Cloud Management to represent all the possible valid combinations of pairs of referencing and referenced CIs in the service definition.
- **Built-in valid relationships:** The following relationships are pre-defined in the base system as valid relationships:
 - cmdb_ci_endpoint -> Applicative Flow To::Applicative Flow From -> cmdb_ci_endpoint
 - cmdb_ci_endpoint -> Implement End Point To::Implement End Point From -> cmdb_ci
 - cmdb_ci -> Use End Point To::Use End Point From -> cmdb_ci_endpoint

General behavior

- Relationship governance rules support inheritance.

For example, suppose that the suggested relationship cmdb_ci_appl Runs On::Runs cmdb_ci_hardware exists. Then a Runs On::Runs relationship between a cmdb_ci_appl_dot_net CI and a cmdb_ci_windows_server CI is valid. That is because .Net Application class inherits from the Application class and the Windows Server class inherits from the Hardware class.

- Duplicate relationship governance rules are not allowed.
- Relationship governance rules are not domain separated.
- It is allowed to have more than one relationship type between the same two CI types.
For example, the following relationships are valid:
 - cmdb_ci_appl Depends On::Used by cmdb_ci_service
 - cmdb_ci_appl Receives data from::Sends data to cmdb_ci_service

Reports

A relationship between CIs is considered valid if it conforms to any of the relationship governance rules. Use the Relationship Health Dashboard to view reports about [relationships health](#) including relationships compliance with relationship governance rules. The 'Relationships not compliant with all relationship rules' report shows CI relationships that are not compliant with any of the relationship governance rules.

Related tasks

- [Add a suggested relationship](#)
- [Create a CI relation rollup](#)

Related concepts

- [CI relations formatter](#)
- [CI relationship editor](#)
- [Relation qualifier](#)
- [CI relationship security](#)

Related reference

- [Suggested class relationships](#)

CI relations formatter

The default CI form includes a CI relations formatter from which you can examine a CI and its relationships in various views. From the CI relations formatter, you can also launch the CI relationship editor for the CI.

If the domain separation plugin is activated, then only relationships in which the logged on user is authorized to view both CIs, are displayed.

The CI relations formatter contains a list of related CIs and a toolbar with controls for viewing the relationships between the current CI and related CIs. You can configure the controls in this formatter to modify varying

aspects of the view. For more information about formatters, see [Create a formatter and add it to the form.](#)

Note:

- If an endpoint is a child in one relationship and the same endpoint is a parent in another relationship, then that endpoint is hidden and does not appear in the relations formatter view. Similarly, relationship qualifier chains are also hidden and do not appear in the relationship formatter view.

- Example: CI1 > endpoint > CI2

In this example, CI1 is related to CI2 through relationships with endpoint. A single relationship appears in the relations formatter:

CI1 > CI2 (These relationships appear as a direct relationship without endpoint, because endpoint is a parent in one relationship and a child in another relationship).

- Example: CI1 > endpoint1 > CI2 > endpoint2

Two relationships appear in the relations formatter:

CI1 > CI2 (endpoint1 is hidden because it is a parent in one relationship and a child in another relationship).

CI1 > CI2 > endpoint2 (appears as level 2 relationship – endpoint1 is hidden and endpoint2 appears as it a child and not a parent in any other relationship).

- On instances that do not meet the internet browser requirements for the CI relations formatter, the default CI form includes the legacy CI relations formatter instead. For more information, see [Legacy CI relations formatter](#).
- CIs not extended from the Configuration Item [cmdb_ci] table, are not displayed in Dependency Views maps and in CI relation formatters.
- The Applicative Flow To::Application Flow From relationship is a special relationship type used only between Service Mapping endpoints. This relationship type is not intended for use in the CMDB as a relationship between CIs and therefore it is not displayed in the relations formatter.

Controls for viewing related CIs

Control	Definition
 Add CI relationship	Starts the relationship editor to manually create CI relationships. For more information (including about roles required for the '+' control to appear), see Create or edit a CI relationship.
 Show dependency views	Launches a Dependency Views map in another window or tab. The CI is the central node in the map, with a configurable number of levels above and below that node in the hierarchy. Map indicators next to the nodes indicate the number of tasks, incidents, problems, changes, or outages related to that node. Right-click to expand collapsed nodes or display a list of related tasks or problems. For more information, see Dependency Views map.
Search for CI	Filters the CIs included in the display.



Click the Settings icon () to configure additional view settings that filter the data displayed. Settings are preserved through logging out and logging back in.

Related items settings

Setting	Description
Show Relations in Flat/Tree Layout	To view a flat list of related CIs that are grouped by relationship

Setting	Description
	type in alphabetical order, click Flat (default value). To view groups of related CIs in a hierarchical tree, click Tree. If you select the tree view, you cannot configure any other settings for viewing related CIs. A single list of upstream and downstream relationships is displayed.
Show Relations in Split/Merge Layout	To view a single list that includes both upstream and downstream relationships, click Merge (default value). Relationships are grouped by relationship type. To view separate lists for upstream and downstream relationships, click Split.
Filter Relations by Max Level	Select the number of downstream and upstream levels in the hierarchy to include when displaying CIs in a flat view. Default value is 3.
Filter Relations by Relationship Type	Select the types of relationships to view. Default value is 'All Relationship Types'.

Setting	Description
Filter Relations by CMDB View	Filter by tables specified in CMDB views, if any relationship filters exist.

The relations formatter uses the following icons to provide additional information about changes, problems, and outages related to CIs in the relationship:

Icons related to CIs

Icon	Description
	Recently closed changes
	Planned changes
	Currently open changes
	Recently closed outages
	Problems
	Incidents
	Planned outages
	Currently open outages

In large networks, a list of related CIs might be excessively long, which can slow performance when a CI form is rendered. You can configure these properties to control the amount of data that is displayed. To find a property, enter `sys_properties.list` in the left navigation filter and search for the property.

Properties related to performance

Property	Description
<code>glide.ecmdb.find_relationship_issues</code>	Hides or displays an icon in the CI relations formatter that links to open issues for the CI. This property defaults to true (displays the icon).
<code>glide.ui.max_relation_levels</code>	Specifies the maximum level for displaying CIs in flat view before reaching the maximum relations limit. The default value is 5.
<code>glide.ui.max_relations</code>	Specifies the maximum number of related CIs to display. When exceeded: - A notification appears indicating that the limit has been reached, and that not all relations are displayed. The default value is 1000. - Flat layout reverts to the tree layout view.

• Domain separation

Domain separation is supported in the relations formatter and the CI relationship editor. Domain separation enables you to separate data, processes, and administrative tasks into logical groupings called domains. You can control several aspects of this separation, including which users can see and access data.

• Create or edit a relationship filter

Create a custom relationship filter to display CI relationships from selected tables in the CI relations formatter.

- [Exclude relationships from the relations formatter view](#)

Create a list of relationships that should not appear in the relations formatter view on CI forms.

- [Legacy CI relations formatter](#)

On instances that do not meet the internet browser requirements for the latest CI relations formatter, the default CI form includes the legacy CI relations formatter instead.

Related tasks

- [Add a suggested relationship](#)
- [Create a CI relation rollup](#)
- [Create or edit a relationship filter](#)

Related concepts

- [Relationship governance rules](#)
- [CI relationship editor](#)
- [Relation qualifier](#)
- [CI relationship security](#)

Related reference

- [Suggested class relationships](#)

Related topics

- [Create a formatter and add it to the form](#)

Domain separation

Domain separation is supported in the relations formatter and the CI relationship editor. Domain separation enables you to separate data, processes, and administrative tasks into logical groupings called domains.

You can control several aspects of this separation, including which users can see and access data.

Overview

How domain separation works in the relations formatter and relationship editor

- **Relations formatter**

The relations formatter is domain-separation supported. The relations formatter is used to display CMDB relationships in the UI in different views. Since the CI Relationship (cmdb_rel_ci) table is not domain separated, relationships are visible in the relations formatter only if both parent and child CIs (cmdb) are visible in the domain.

The CI Relationship Type (cmdb_rel_type) table is not domain separated. Therefore, in the relations formatter, all the relationship types are available to be selected as a filter.

By default domain separation is supported in the relations formatter.

- **Relationship editor**

The relationship editor is domain-separation supported. You can use the relationship editor to add new relationships or delete existing relationships for the current CI.

- The CI relationship editor displays a list of CIs to add or remove from relationships. Since they are domain separated, the CI list view in the relationship editor displays the CIs that are visible to the current domain.
- The CI relationship editor displays a list of relationships to add or remove. Since the CI Relationship [cmdb_rel_ci] table is not domain separated, the relationships list view displays all the relationships of the current CI.

The Suggested Relationship (cmdb_rel_type_suggest) table is not domain separated, which means that all the suggested relationship types in the relationship editor are visible for all domains.

By default domain separation is supported in the relationship editor.

Related concepts

- [Domain separation and Configuration Management Database \(CMDB\)](#)

Create or edit a relationship filter

Create a custom relationship filter to display CI relationships from selected tables in the CI relations formatter.

Before you begin

Role required: `ecmdb_admin`

About this task

The CI relations formatter displays related CIs for the base CI, and the relationships between the CIs. You can use relationship filters on the CI relations formatter to customize CI relationship views.

Procedure

1. Navigate to **All > Configuration > Relationships > Relationship Filters**.
2. Click **New** or select a filter to edit.
3. Enter or edit the relationship filter name.
4. Right-click the form header and click **Save**.
5. In the **Configuration Types** section, click **Edit**.
6. On the **Edit Members** form, select the tables of the CIs that you want to show with the filter and then move the tables to the **Configuration Types list**.
7. Click **Save**.

Result

On a CI form, in the relations formatter settings, you can select the newly defined relationship filter from the **Filter Relations by CMDB View** list.

In the legacy CI relations formatter, you can click **View** and select the newly defined relationship filter.

After you select a filter, the relations formatter displays only CIs from the tables specified in the filter or from descending tables.

Exclude relationships from the relations formatter view

Create a list of relationships that should not appear in the relations formatter view on CI forms.

Before you begin

Role required:

- To view the relationship type exclusion list — itil
- To create, update, or delete the relationship type exclusion list — itil_admin

Procedure

1. Navigate to **All > Configuration > Relationships > Relationship Type Exclusion List**.
2. In the CI Relation Filters list view, click **New**.
3. Fill out the CI Relation Filter form to specify the relationship that you want to exclude from view.
4. Click **Submit**.
Excluded relationships do not appear in the relations formatter view on CI forms.

Legacy CI relations formatter

On instances that do not meet the internet browser requirements for the latest CI relations formatter, the default CI form includes the legacy CI relations formatter instead.

This element contains the list of related CIs and a toolbar with controls for viewing the relationships between the current CI and related CIs.

For information about the latest CI relations formatter, see [CI relations formatter](#).

Related items field



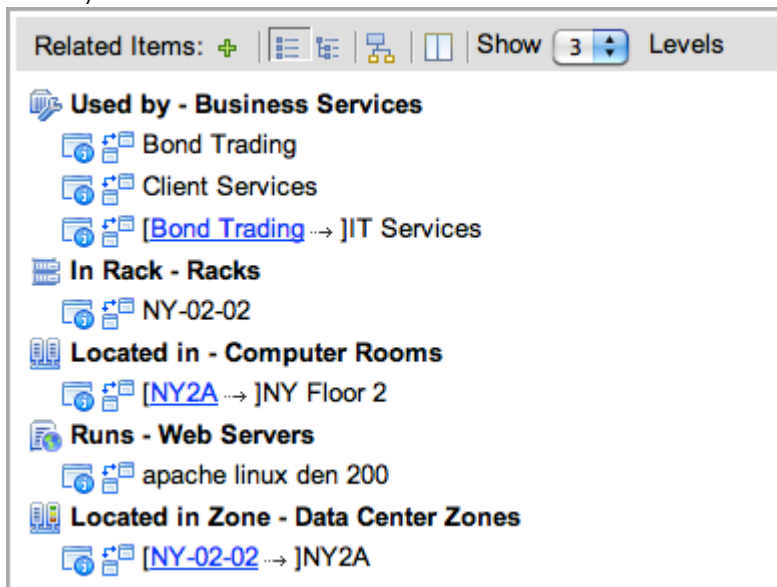
Note: The legacy BSM map provides a more complete view of CI relationships.

Configure the controls in this formatter with two properties that restrict varying aspects of the view.

Flat layout

Click the flat layout icon () to group related CIs by relationship.

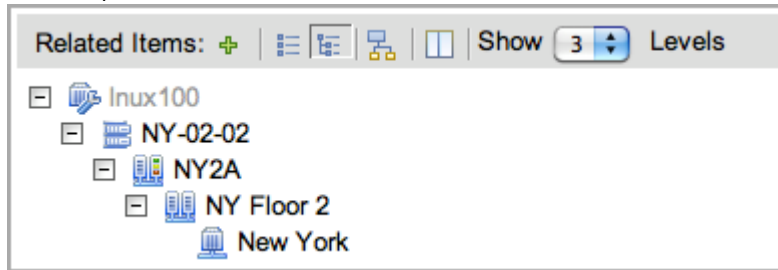
Flat layout view



Tree layout

Click the tree layout icon () to group related CIs in a hierarchical tree.

Tree Layout view



CI relationship editor

Use the relationship editor to view, create, modify, or delete CI relationships. Open the relationship editor from the CI Relations formatter.

As an alternative to the CI relationship editor, [Unified Map](#) in the [CMDB Workspace store app](#) provides the latest functionality for editing CI relationships. For more information, see [Edit relationships in Unified Map](#).

When you use the relationship editor, the CI from which the editor was launched is designated as the base CI. You can then select one or more CIs as a second CI for the relationship. Depending on the selected relationship type, the base CI can become the parent CI or the child CI in the new relationship.

The relationship editor operates differently, depending on whether you select the **Use suggested relationship** check box.

- With suggested relationships, the relationship editor lists all available relationship types for the base CI. To define a new relationship, select a relationship type, and then select a second CI for the relationship.

Suggested relationships are highlighted for you. These relationships are displayed in blue with a prefix of [Suggested].

- Without suggested relationships, you define a new relationship by first selecting a second CI for the relationship and then selecting a parent or a child relationship type.

Note: The following relationship types are used only for Service Mapping endpoints, and you cannot use them as a relationship type between two CIs:

- Implement End Point To:Implement End Point From
- Use End Point To: Use End Point From
- Applicative Flow To:Applicative Flow From

Suggested relationships

If you select the **Use suggested relationship** check box in the editor, the **Suggested relationship** list appears. It displays all available CI, user and group relationship types for the base CI. Relationship types have a suffix of **(Parent)** or **(Child)** to note the relationship descriptor, and suggested relationship types are displayed in blue and have a "*" prefix.

When you select a relationship, you are also designating the base CI as being the parent or the child CI in the new relationship. For example, if you select the 'Feeds' relationship type, the base CI becomes the designated parent CI, and the second CI that you select becomes the child CI in this relationship.

Downstream relationships

If you do not select the **Use suggested relationship** check box in the editor, the **Downstream relationships** list appears. It displays all relationships in which the base CI is the parent CI. The child CI of the relationship is displayed in the **Child** column.

Upstream relationships

If you do not select the **Use suggested relationship** check box in the editor, the **Upstream relationships** list appears. It displays all relationships in which the base CI is the child CI. The parent CI in each relationship is displayed in the **Parent** column.

- [Create or edit a CI relationship](#)

Use the relationship editor to view, create, or modify CI relationships. You can open the relationship editor from the CI Relations formatter.

- [Delete a CI relationship](#)

Maintain the integrity of the CMDB by deleting any CI relationships that are no longer relevant or needed for a CI. Use the relationship editor to delete CI relationships.

- [Legacy CI relationship builder](#)

Used to define CI relationships manually, this page is a sophisticated version of the standard list collector. In the legacy CI relations formatter, click the CI relationship builder icon () to display the legacy Define Relationships page.

Related tasks

- [Add a suggested relationship](#)
- [Create a CI relation rollup](#)
- [Add a suggested relationship](#)
- [Create or edit a CI relationship](#)

Related concepts

- [Relationship governance rules](#)
- [CI relations formatter](#)
- [Relation qualifier](#)
- [CI relationship security](#)
- [Legacy CI relationship builder](#)

Related reference

- [Suggested class relationships](#)

Create or edit a CI relationship

Use the relationship editor to view, create, or modify CI relationships. You can open the relationship editor from the CI Relations formatter.

About this task

As an alternative to the CI relationship editor, [Unified Map](#) in the [CMDB Workspace store app](#) provides the latest functionality for editing CI relationships. For more information, see [Edit relationships in Unified Map](#).

Before you begin

Role required:

- To create relationships: ITIL or asset
- To view relationships, depending on the state of the Table API ACL:
 - If inactive (default): ITIL or asset
 - If active: ITIL or asset, and `snc_platform_rest_api_access`

For more information, see [CI relationship security](#), [REST API](#), and [Table API](#).

About this task

The relationship editor operates differently, depending on whether you check the **Use suggested relationship** option or not.

Procedure

1. Launch the relationship editor:
 - a. Open a CI form.
For example:
 - a. Navigate to **All > Configuration > CI Class Manager**.
 - b. On the CI Class Manager page, select **Open Hierarchy**.
 - c. On the CI Class panel search and select the class of the CI for which you want to create or edit relationships.
 - d. Select **CI List** in the left-side panel and then in the list view, select the CI.
 - b. On the CI form, scroll down as needed to select the Add CI relationship icon (+ sign) in the Related Items section.

2. To use suggested relationships, first select a relationship type, and then select one or more CIs to be the child CIs in the relationship:

Suggested relationship types

☒ Use suggested relationships

Filter

Class: is a Computer AND OR X

Location: is anything AND OR X

Operational status: is anything AND OR X

Name: starts with *Carol AND OR X

Configuration Items

Name	Location	Description	Class	Updated	Maintenance schedule
*CAROL1-IBM	Lenovo	322 West 52nd Street, New York, NY	Computer	2017-10-25 02:53:26	
*CAROL2-IBM	Lenovo	322 West 52nd Street, New York, NY	Computer	2017-10-25 02:53:12	
*CAROL3-GATEWAY	Gateway	322 West 52nd Street, New York, NY	Computer	2017-10-25 02:52:53	

Relationships

Type	Parent	Child
Depends on:Used by	abehermandez-snc-2017-10-10070721-buildtools1	*CAROL1-IBM
Depends on:Used by	abehermandez-snc-2017-10-10070721-buildtools1	*CAROL2-IBM
Depends on:Used by	abehermandez-snc-2017-10-10070721-buildtools1	*CAROL3-IBM

Newly created relationships

[Click to add a new relationship](#)

- a. Select **Use suggested relationship**.
- b. From the **Suggested relationship type** list, select a relationship type.
You can filter the list of suggested relationships by using the filter check boxes.

Filter option	Description
Hide CI relationship	Hides any relationships between the base CI and another CI (such as "Receives data from"). Default filter is stored in the ci_manage_relationships_filter_hint.cmdb_ci user preference.
Hide user relationship	Hides any relationships between the base CI and a user (such as "Logs reviewed by"). The default filter is stored

Filter option	Description
	in the ci_manage_relationships_filter_ hint.sys_user preference.
Hide group relationship	Hides any relationships between the base CI and a group (such as "Backups done by"). Default filter is stored in the ci_manage_relationships_filter_ hint.sys_user_group user preference.

The **Configurations Items** list displays all the CIs that are appropriate for the base CI and the selected relationship type. The **Relationships** list at the bottom of the editor, displays all existing relationships of the selected relationship type, in which the base CI is a parent CI or a child CI.

- c. From the **Configuration Items** list, select one or more CIs as a second CI for the relationship.
You can filter the list of **Configurations Items** by adding conditions in the **Filter** section and clicking **Run filter**.

If you selected a parent relationship type, these CIs becomes the child CI in the relationship, and if you selected a child relationship type, then the selected CIs become the parent CI in the relationship.

- d. In the **Relationships** section, click the plus icon (+) to add the new relationships.
Alternatively, you can drag the selected CIs to the **Relationships** list. Each new relationship will consist of the base CI, the selected relationship type, and a selected second CI.

3. To not use suggested relationships, first select one or more CIs to be the child CIs in the relationship, and then select the relationship type:

The screenshot shows the ServiceNow Configuration Items interface. At the top, there is a checkbox labeled "Use suggested relationships" with a red callout bubble saying "Clear to not use suggested relationships". Below this is a "Filter" section with various dropdowns and buttons. A red callout bubble points to the "Run filter" button, stating "Filtered list of CIs appropriate for the base CI and selected relationship type". The main section is titled "Configuration Items" and contains a table with columns: Name, Location, Description, Class, Updated, and Maintenance schedule. The table lists three items: *CAROL-IBM, *CAROL2-IBM, and *CAROL3-GATEWAY. Below the table are two sections: "Downstream Relationship" and "Upstream Relationship". A red callout bubble points to the "+" button in the "Downstream Relationship" section, saying "Click to add downstream new relationship". Another red callout bubble points to the "+" button in the "Upstream Relationship" section, saying "Click to add upstream new relationship". A dropdown menu is open in the "Upstream Relationship" section, showing options: "Please Select a Relationship", "None", "Member of", "Powered by", "Runs", and "Used by". A red callout bubble points to this dropdown, saying "Newly created relationships". At the bottom, there are "Save and Exit", "Save", and "Cancel" buttons.

- a. Clear **Use suggested relationship**.
- b. In the **Configuration Items** list, select one or more CIs as a second CI for the relationship.
You can filter the list of **Configurations Items** by adding conditions in the **Filter** area and clicking **Run filter**. Depending on the relationship type that you will select, the selected CIs might become a parent or a child CI in the relationship.
- c. With at least one CI selected in the **Configuration Items** list, click the '+' sign in the **Downstream Relationships** section or the **Upstream Relationships** section to create the relationship.
 - Add the relationship to **Downstream Relationships** to create a relationship in which the base CI is the parent CI and the selected CI is the child CI.
 - Add the relationship to **Upstream Relationships** to create a relationship in which the base CI is the child CI and the selected CI is the parent CI.
- d. For each newly created relationship in either the **Downstream Relationships** or the **Upstream Relationships** lists, click **Please select a relationship** and select a relationship type.
 - The list of available relationship types in the **Downstream Relationships** list contains parent relationships only, in which the base CI is the parent CI.

- The list of available relationship types in the **Upstream Relationships** list contains child relationships only, in which the base CI is the child CI.
- e. Click **Save** or **Save and Exit**.
Only after you enter all the information that is necessary for creating the relationship, these buttons light up indicating that there are pending updates that require saving.

Related concepts

- [CI relationships in the CMDB](#)
- [CI relationship security](#)

Related reference

- [Suggested class relationships](#)

Delete a CI relationship

Maintain the integrity of the CMDB by deleting any CI relationships that are no longer relevant or needed for a CI. Use the relationship editor to delete CI relationships.

About this task

As an alternative to the CI relationship editor, [Unified Map](#) in the [CMDB Workspace store app](#) provides the latest functionality for editing CI relationships. For more information, see [Edit relationships in Unified Map](#).

Before you begin

Role required: ITIL or asset

Note: Deleting a relationship to a dependent CI can result in identification problems as the dependent CI will no longer have a relationship to the CI it depends on.

Procedure

1. Launch the relationship editor:
 - a. Open the CI form of the CI for which you want to delete a relationship.
 - b. Locate the **Related Items** section near the center of the form.
 - c. Click the plus (+) icon on the **Related items** section.
2. In the Relationship Editor, in the Relationships section, select the relationships that you want to delete for the CI.
3. Click the **Delete selected relationships** icon.
4. Click **Save** or **Save and Exit**.

Legacy CI relationship builder

Used to define CI relationships manually, this page is a sophisticated version of the standard list collector. In the legacy CI relations formatter, click the CI relationship builder icon () to display the legacy Define Relationships page.

For information about the latest CI relationship editor, see [CI relationship editor](#).

Select a CI relationship type

The top half of the legacy relationship editor contains a large option box that allows you to select which type of relationship you want to manipulate. Click the particular type of relationship you are interested in working with.

Filter the list of CI relationships

In the legacy relationship editor, the check boxes along the right hand edge of the select box provide a quick way to filter down the list of available relationships.

By default, the system displays a list of all suggested relationships for the type of CI you selected. For example, if you selected a Database instance, a relationship of "Runs on" makes sense, but a relationship of

"Provides HVAC for" does not. The default filter is stored in the user preferences

ci_manage_relationships_filter_hint.cmdb_cici_manage_relationships_filter_hint.sys_user, and ci_manage_relationships_filter_hint.sys_user_group.

- **Hide CI relationship** -- Hides any relationships between this CI and another CI (e.g. "Receives data from").
- **Hide user relationships** -- Hides any relationships between this CI and a user (e.g. "Logs reviewed by").
- **Hide group relationships** -- Hides any relationships between this CI and a group (e.g. "Backups done by").
- **Show all relationships** -- If you have the appropriate role (out of the box this is itil_admin) you will have an additional check box labeled "Show all relationships." If you click that check box, the system will let you choose any relationship defined in the system, regardless of where it is on the "suggested" list for this type of CI.

Select CI relationship targets

In the legacy relationship editor, users can link or unlink CIs for a relationship type.

As soon as you pick a relationship type, the system will fill in the two select boxes at the bottom of the screen with CI's that are appropriate for the relationship you suggested. The left hand select box will contain a list of CI's that might reasonably be linked via this relationship, while the right hand box contains a list of those CI's which are already linked.

1. Link or unlink items.

Link new items	Move that CI from the left hand box to the right hand box.
Unlink existing items	Move them from the right hand box to the left.

When you make either type of change, a message appears indicating that you have pending changes.

2. Apply or cancel your changes.

Click the Save button.	This will save your set of changes, and go back to the previous screen (either a CI or the BSM map depending on how you got here).
Click the Cancel button.	This causes you to exit without saving your changes.

Related concepts

- [CI relationships in the CMDB](#)

Relation qualifier

A relation qualifier, which is a CI of the Qualifier [cmdb_ci_qualifier] type, stores important information about the CI relationships.

In a relation qualifier, you can annotate arbitrary unique information about the relationship between two CIs. You can define multiple qualifiers for a single relationship, resulting in a qualifier chain. But, there can be only a single qualifier chain for a specific relationship type between two CIs.

For example, for a relationship between a parent CI and a child CI, you can add a relation qualifier to note that the relationship was discovered based on traffic (such as cmdb_ci_qualifier_traffic_based_connection). This results in having two records in the CI Relationship [cmdb_rel_ci] table for the relationship.

- A record that links the parent CI and the new qualifier
- A record that links the new qualifier and the child CI

For this relationship, there is a parent CI and a child CI, and a relation qualifier of type cmdb_ci_qualifier_trafficbased.

For information about usage of relation qualifiers in the identification process, see [Identification rules](#).

Related tasks

- [Add a suggested relationship](#)
- [Create a CI relation rollup](#)

Related concepts

- [Relationship governance rules](#)
- [CI relations formatter](#)
- [CI relationship editor](#)
- [CI relationship security](#)

Related reference

- [Suggested class relationships](#)

CI relationship security

When applying security to CI relationships, it is important to apply the access controls both to the CI Relationship (cmdb_rel_ci) table and to create an operation editCIRelations to the * table as well.

If the current instance has defined security for editCIRelations, it will be applied to edit_ci_relations automatically in the process of upgrading, and the out-of-date security will be removed.

Related tasks

- [Add a suggested relationship](#)
- [Create a CI relation rollup](#)
- [Create a CI relation rollup](#)

Related concepts

- [Relationship governance rules](#)

- CI relations formatter
- CI relationship editor
- Relation qualifier
- CI relationships in the CMDB

Related reference

- [Suggested class relationships](#)

Create a CI relation rollup

A CI relation rollup allows you to sum, count, max, min, or mean a relationship type. You can create CI relation rollups.

Before you begin

Role required: `ecmdb_admin`

About this task

CI relation rollup can be useful for tracking and for receiving notifications. For example:

- In a sum roll up, add up fields from multiple CIs and display the result on another CI to which they are related. So, if you have four configuration items in a rack that are all consuming power, create a CI relation rollup to add all the power usage together and display the result in one field on the rack CI form.
- If a certain level of power consumption in a rack is exceeded, send a notification.
- With a rack that has 10 slots, send a notification when 9 slots are filled.

CI relation rollups use the `cmdb synch event` business rule on the `[cmdb_ci]` table. Although this business rule is active by default, you must modify the rule slightly before it will run.

Procedure

1. Navigate to **All > Configuration > Relationships > CI Relation Rollups**.
2. Click **New**.
3. Complete the form.

CI Relationship Rollup fields

Field	Description
CI Relationship Type	Select a relationship type from the list to use with the rollup. For example, Members::Member of contains the parent descriptor Members and the child descriptor Member of .
Type	Select the type of rollup from the drop-down list: COUNT, MAX, MEAN, MIN, or SUM.
Parent field	The target field on which the operation will be done.
Child field	The input to the equation type. The Parent field is affected by the selections in the child field.
Rollup class	The classes that can use the relationship. For example, you can specify that the relationship only applies to racks.

4. To run the cmdb synch event business rule, navigate to **Business Rules**.
5. Use the search box to find the [cmdb synch event] table.
6. Click the cmdb synch event business rule to go to the **Business Rule** page.
7. Select the **Update**, **Delete**, and **Query** check boxes.

Additionally, if you wish CI relation rollups to recalculate when there is a change to a relationship, use a similar procedure to select the **Active** check box on the cmdb_rel_ci synch event business rule.

Related tasks

- [Add a suggested relationship](#)

Related concepts

- [Relationship governance rules](#)
- [CI relations formatter](#)
- [CI relationship editor](#)
- [Relation qualifier](#)
- [CI relationship security](#)
- [CI relationships in the CMDB](#)

Related reference

- [Suggested class relationships](#)